

Disproof of conjecture 00000000463

earthking11

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Verdict: FALSE

We disprove conjecture 00000000463 as stated. The refutation is unconditional and completely elementary: for every $n \geq 4$ the integer n^{n-2} fails to be squarefree, so the counting function on the left is bounded, whereas $cN/\sqrt{\log N}$ tends to infinity for every $c > 0$. The claimed asymptotic equivalence cannot hold.

1 The conjecture, quoted verbatim

From `conjectures/00000000463.md`:

English. Definition: $\tau(K_n) = n^{n-2}$. Conjecture: $\#\{n \leq N : n^{n-2} \text{ squarefree}\} \sim c \cdot N/\sqrt{\log N}$ (a density compatible with abc).

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim (in Unicode) in `README.md`, and the two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

Here $\tau(K_n) = n^{n-2}$ is Cayley's formula for the number of spanning trees of the complete graph K_n . The conjecture asserts that the set of $n \leq N$ for which this tree count is squarefree has asymptotic size $cN/\sqrt{\log N}$ for some constant c .

2 Conventions

The statement is informal at three points; we fix the conventions explicitly and show that the verdict is the same under every reasonable choice.

1. **Domain of n .** We take $n \in \{1, 2, 3, \dots\}$, as the notation $\#\{n \leq N\}$ suggests.
2. **The exponent $n - 2$ at $n = 1$.** $1^{1-2} = 1^{-1}$ is not defined in $\mathbb{N}$ (negative exponents leave $\mathbb{N}$); it is 1 in $\mathbb{Z}$ or $\mathbb{Q}$. We therefore *exclude* $n = 1$ from the main count and record separately that including it (with the value $1^{-1} := 1$) adds exactly one index.
3. **Is 1 squarefree?** Yes, by the standard convention: an integer is squarefree when no prime square divides it, and no prime divides 1 (equivalently, its factorisation is empty). Thus $2^0 = 1$ and, under the convention of item 2, $1^{-1} = 1$ are squarefree, while 0 is not.
4. **Meaning of $\sim$.** The standard meaning in analytic number theory: $f(N) \sim g(N)$ means $f(N)/g(N) \rightarrow 1$ as $N \rightarrow \infty$. In particular g must be unbounded when f is.

3 The argument

Theorem 1. *For every integer $n \geq 4$, n^{n-2} is not squarefree. Consequently*

$$\#\{n \leq N : n^{n-2} \text{ squarefree}\} = \begin{cases} 2, & N \geq 3 \text{ (counting } n \geq 2), \\ 3, & N \geq 3 \text{ (counting } n \geq 1 \text{ with } 1^{-1} := 1), \end{cases}$$

a bounded function of N .

Proof. Let $n \geq 4$ and let p be any prime divisor of n ; such a p exists because $n \geq 2$. Write $v_p(\cdot)$ for the p -adic valuation. Since $n^{n-2} = \prod_q q^{(n-2)v_q(n)}$, we get

$$v_p(n^{n-2}) = (n-2)v_p(n) \geq 2 \cdot 1 = 2.$$

Hence $p^2 \mid n^{n-2}$: a prime square divides n^{n-2} , so it is not squarefree.

For the count: for $n \geq 4$ nothing contributes. For $n = 2$ we have $2^0 = 1$, which is squarefree; for $n = 3$ we have $3^1 = 3$, which is squarefree. Under the convention $1^{-1} := 1$ the value $n = 1$ is squarefree as well. No other n contributes, so the count is 2 for $N \geq 3$ when $n \geq 2$, and 3 when $n \geq 1$ is included with that convention. $\square$

Corollary 2. *For every real $c > 0$, $\#\{n \leq N : n^{n-2} \text{ squarefree}\}$ is not asymptotic to $cN/\sqrt{\log N}$.*

Proof. The left-hand side is bounded by 3 for all N (Theorem 1) while $cN/\sqrt{\log N} \rightarrow \infty$ for every $c > 0$. Therefore

$$\frac{\#\{\dots\}}{cN/\sqrt{\log N}} \leq \frac{3\sqrt{\log N}}{cN} \rightarrow 0 \neq 1,$$

so the defining ratio for $\sim$ does not tend to 1. (If $c = 0$ no asymptotic equivalence is defined, and the parenthetical “a density compatible with abc” clearly intends $c > 0$.) $\square$

Remark 3. The failure is not a matter of the constant c : no choice of c can make a bounded sequence asymptotic to a sequence that diverges. It is also not a matter of the domain of n ; see Section 5.

4 The count table

The table compares the true count with the conjectured main term; we use $c = 1$, and note that any $c > 0$ only changes the right-hand column by the fixed factor c . Both normalisations of the count (with and without $n = 1$) are shown.

N	$\#\{n \leq N : n \geq 2, n^{n-2} \text{ sqfree}\}$	$\#\{n \leq N : n \geq 1, n^{n-2} \text{ sqfree}\}$	$N/\sqrt{\log N}$
10	2	3	6.59
10^2	2	3	46.60
10^3	2	3	380.48
10^4	2	3	3295.05
10^5	2	3	29471.83
10^6	2	3	269039.80

The only squarefree values among $2 \leq n \leq 3000$ are

$$n = 2 : 2^0 = 1, \quad n = 3 : 3^1 = 3,$$

verified by exact trial division / factorisation. The left columns are constant, the right column grows without bound.

5 Alternative readings

We tried to save the conjecture by changing each convention; none succeeds.

Reading	Outcome
n starts at 2	count is exactly 2 for all $N \geq 3$; still bounded, still refuted.
n starts at 1 with $1^{-1} := 1$ and 1 square-free	count is 3 for all $N \geq 3$; still bounded, still refuted.
“squarefree” excludes 1	then $n = 2$ ($2^0 = 1$) is also excluded and the count is 1; even more bounded.
Domain restricted to primes n	for prime $n \geq 5$, $v_n(n^{n-2}) = n - 2 \geq 3$, so not squarefree; only $n = 2, 3$ survive, count 2; refuted.
Read “ n^{n-2} squarefree” as “ n square-free”	then the count is $\sim (6/\pi^2)N$, whose ratio to $cN/\sqrt{\log N}$ is $\sim (6/(\pi^2 c))\sqrt{\log N} \rightarrow \infty$, so no constant c works; refuted.
Read “ $\sim$ ” as “ $\ll$ ” (big- O)	a bounded count is $O(N/\sqrt{\log N})$, but this is a different (weaker) claim and contradicts both the standard meaning of $\sim$ and the phrase “a density compatible with abc”. Under the literal, standard reading the claim is false.

Only the last line changes the meaning of a symbol rather than a convention on the domain; it does not express the conjecture as filed, so the refutation stands. We record it as a caveat, not as an ambiguity.

6 Reproducibility

The exhaustive check is carried out by `reproduce.py` using the Python 3 standard library only (no `sympy`):

```
$ python3 reproduce.py
... (exact factorisation of the base n, exhaustive for 2 ≤ n ≤ 3000, the
count table, and the checks)
PASS: all checks verified; conjecture 00000000463 is FALSE.
```

It factors each base n by trial division and uses the exact identity $v_p(n^{n-2}) = (n-2)v_p(n)$; for $n \leq 12$ it additionally factors n^{n-2} directly by trial division and confirms agreement. It prints `PASS` and exits 0 exactly when every check holds.

7 Formalisation

The refutation is formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). The main statements are

```
theorem not_squarefree_pow (n : Nat) (h : 4 ≤ n) :  
  ¬ Squarefree (n ^ (n - 2))  
theorem squarefree_one : Squarefree 1  
theorem squarefree_two_pow : Squarefree (2 ^ (2 - 2))  
theorem squarefree_three_pow : Squarefree (3 ^ (3 - 2))  
theorem sqfreeCount_le_three (N : Nat) : sqfreeCount N ≤ 3
```

where `Squarefree m` is $\forall p, \text{Prime } p \rightarrow \neg (p \cdot p \mid m)$ and `Prime` is primality as a `Prop` (since `Nat.Prime` is not available in `import Std`). The proof of `not_squarefree_pow` uses the existence of a prime divisor of n (proved by strong induction) and the divisibility step $p \mid n \Rightarrow p^2 \mid n^{n-2}$. The count function is proved to be $\min N3$. See `lean4/README.md` for the full statement table, the axiom audit, and the scope note.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 and the audit reports no `sorryAx`.